S21 — Stimulus quality x word frequency on ex-Gaussian mu

Question. In lexical decision with pseudohomophone nonwords, is the frequency effect (low- minus high-frequency) on the ex-Gaussian mu of correct word RTs larger for degraded than for clear words?

Data and sample. Replication file `d07.csv`: 71 participants, 22,481 trials. The file already contains only correct trials with RTs between 200 and 3,000 ms (all accuracy codes = 1, RT range 203–2,998 ms), so the error and 200/3,000 ms exclusions were applied before the file was shared; how many trials they removed cannot be recovered from it. I then applied the 2.5-SD trim as specified (per participant, separately for word and nonword trials): it removed 598 trials (359 word, 239 nonword). Analysed: 11,457 correct word trials, 71 participants (none excluded), 18–50 trials per participant x condition cell (median 41). Frequency coding: `ldtfreq` level 1 has more correct trials from an equal number of items (higher accuracy) and was taken as high frequency; it is also faster overall, consistent with that.

Measure. For each participant x condition, an ex-Gaussian was fitted to the RTs by quantile maximum likelihood (my own implementation: likelihood of the counts between the 10th, 20th, … 90th percentiles of the data; QMPE v2.18 itself was not available). mu is the analysed value. All 284 fits converged.

Primary test (pre-specified). 2 (quality) x 2 (frequency) repeated-measures ANOVA on mu; key term the quality x frequency interaction, F(1, 70); partial eta^2 = SS_int / (SS_int + SS_error).

mu (ms), mean over 71 participants	High frequency	Low frequency	Frequency effect
Clear	560.8	578.5	17.7
Degraded	679.2	783.8	104.6

Result. Interaction: F(1, 70) = 11.41, p = .0012; partial eta^2 = .140, 95% CI [.024, .289]. The interaction contrast (LF−HF)degraded − (LF−HF)clear = +86.9 ms, 95% CI [35.6, 138.2], in the predicted direction. Main effects: quality F(1, 70) = 22.07, p < .001 (degraded slower by 162 ms on mu); frequency F(1, 70) = 21.90, p < .001 (61 ms).

Robustness. (a) mu by continuous maximum likelihood: interaction +98.4 ms, F(1, 70) = 13.17, p = .0005, partial eta^2 = .158. (b) Without the 2.5-SD trim, QMLE mu: +84.3 ms, F(1, 70) = 9.83, p = .0025, partial eta^2 = .123. (c) Not the key test: on mean RT the interaction is absent and slightly in the other direction, −13.3 ms [−37.0, 10.3], F(1, 70) = 1.26, p = .26 — the larger degraded-word frequency effect appears in the leading edge (mu), not in the means.

Caveats. Fits use about 40 trials per cell, so individual mu estimates are noisy (wide participant-level spread in the quality effect); my QMLE is not QMPE v2.18 and its quantile set

may differ, but two estimators and both trimming choices agree. The upstream exclusions could not be audited.

Conclusion. The claim holds on this data: the frequency effect on ex-Gaussian mu is larger for degraded than for clear words (interaction F(1, 70) = 11.41, p = .0012, partial eta^2 = .14).

Files: `analysis.py` (run from the S21 directory; `--no-trim` for robustness b), `analysis.out`, `analysis_notrim.out`, `cell_estimates.csv`, `log.md`.